SAINT LOUIS UNIVERSITY
School of Accountancy, Management, Computing and Information Studies

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Maryheights Campus, Bakakeng, 2600, Baguio City, Philippines

DETECTING ACTIONABLE SUGGESTIONS IN FILIPINO-ENGLISH CODE-SWITCHED GOVERNMENT SOCIAL MEDIA COMMENTS

A Requirement in the Course
CSE 30 - 9712 Special Topics 1
1:30-3:00 MTh D512

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. . → tentative, subject to change from further research and model development
. . → notes

I. Abstract

II. Introduction

Background of the Study

Social media provides a channel through which citizens can express concerns and communicate feedback to government agencies. In the Philippines, researchers have used social media comments to analyze public sentiment toward government agencies and policies. Umali and Miranda (2020) examined comments concerning selected Philippine government agencies, while Vizmanos et al. (2023) analyzed public sentiment toward government policies using comments on the Philippine Institute for Development Studies' Facebook posts.

However, identifying public sentiment does not necessarily identify the action a citizen wants the government to take. For example, "Ang pangit ng kalsada namin" expresses dissatisfaction, whereas "Please repair the damaged road in our barangay" requests a specific action. This distinction is important because a system that identifies actionable suggestions and requests must recognize what citizens propose or ask the government to do, rather than simply classify their opinions as positive, negative, or neutral.

The task also involves Filipino-English code-switching, in which speakers alternate between Filipino and English within a conversation or utterance (Bautista, 2004). Comments such as "Dapat ayusin ang drainage" and "Please po, clean the canal" illustrate how requests may appear in either language or a mixture of both. A classification model must therefore be evaluated on comments containing these language patterns.

Problem statement and Research gap

Philippine studies have applied sentiment analysis to government-related social media comments (Umali & Miranda, 2020; Vizmanos et al., 2023). Research on speech-act classification has also examined how language expresses intentions related to actions (Laurenti et al., 2022). However, these studies do not establish whether adding suggestion-related linguistic features to a TF-IDF

and Logistic Regression model improves the detection of actionable suggestions and requests in Filipino-English code-switched comments from Philippine national government Facebook pages.

This study addresses that specific research problem by comparing a text-based baseline model with a model that incorporates additional suggestion-related linguistic features. The research gap is limited to this classification task and model comparison; the extent of the gap will be established through the study's systematic review of related literature.

Objectives

The general objective is to develop and evaluate a machine-learning model for detecting actionable suggestions and requests in Filipino-English code-switched comments on selected Philippine national government Facebook pages.

The specific objectives are to:

1. Collect and manually annotate eligible public Facebook comments into two classes: actionable suggestion/request and other comment.
2. Develop a baseline model using Term Frequency–Inverse Document Frequency (TF-IDF) and Logistic Regression.
3. Develop a second model using Logistic Regression with TF-IDF and additional suggestion-related linguistic features, including "please," "dapat," "sana," and "paki-."
4. Compare the models using precision, recall, F1-score, macro-F1, and a confusion matrix on the same held-out test dataset.

Research question

Does incorporating suggestion-related linguistic features improve the detection of actionable suggestions and requests in Filipino-English code-switched government social media comments compared with a TF-IDF and Logistic Regression baseline?

Significance

This study will provide an empirical comparison of two classification approaches for identifying actionable suggestions and requests in Filipino-English code-switched government social media comments. Its labeled dataset, annotation guidelines, and evaluation results may support future research on Filipino-English natural language processing and automated civic feedback analysis.

The proposed model may also serve as a preliminary tool for organizing citizen comments according to whether they contain actionable suggestions or requests. However, this study will evaluate classification performance only; it will not measure government responsiveness or determine whether identified suggestions lead to policy changes or improved public services.

Scope

The study will examine public Facebook comments on selected Philippine national government agency pages. It will include comments written in Filipino, English, or a combination of both. The target dataset will consist of <number> usable comments, subject to data accessibility, collection permissions, and the inclusion criteria established in the methodology.

Each comment will receive one of two labels. Actionable suggestion/request refers to a comment that proposes an identifiable action or asks for a specific action or response. Other comment refers to a comment that does not meet this definition. Complaints without a proposed action and general wishes without an identifiable action will be classified as other comments under the annotation guidelines.

The experiment will compare a TF-IDF and Logistic Regression baseline against the same classifier augmented with predefined suggestion-related linguistic features. Both models will use the same held-out test dataset and evaluation metrics.

The study will not classify sentiment as its primary task, verify the factual accuracy of comments, assess the feasibility of proposed actions, or measure government responses. Its findings will apply to the collected dataset and selected pages, not automatically to all Philippine government agencies or Filipino social media users.

III. Review of Related Literature and Studies

A. Government Social Media Sentiment Analysis
B. Actionability and Speech-Act Detection
C. Code-Switching and Linguistic Features

Specific research gap

IV. Methodology

Research Design and Dataset: Where the comments come from, collection period, inclusion criteria, and intended dataset size.
Annotation and Preprocessing: How comments are labeled as actionable suggestions/requests or other comments, how disagreements are handled, and how text is prepared.
Baseline Model: TF-IDF + Logistic Regression.
Proposed Model: The same classifier with additional suggestion-related linguistic features.
Experimental Setup and Evaluation: Train/test strategy, precision, recall, F1-score, macro-F1, and the model-comparison procedure.

V. Results and Discussion

Describe the final dataset and the number of comments in each class.
Present a comparison table for the baseline and proposed model.
Explain whether the additional features improved performance.
Discuss important false positives and false negatives.
Compare the findings with relevant prior studies.
State whether the results support your hypothesis, if one was specified.

VI. Conclusion and Recommendations

Summarization, answer the research question, limitation/s, recommendation/s.

References
Appendices